

Cardiomyopathies are heart muscle diseases

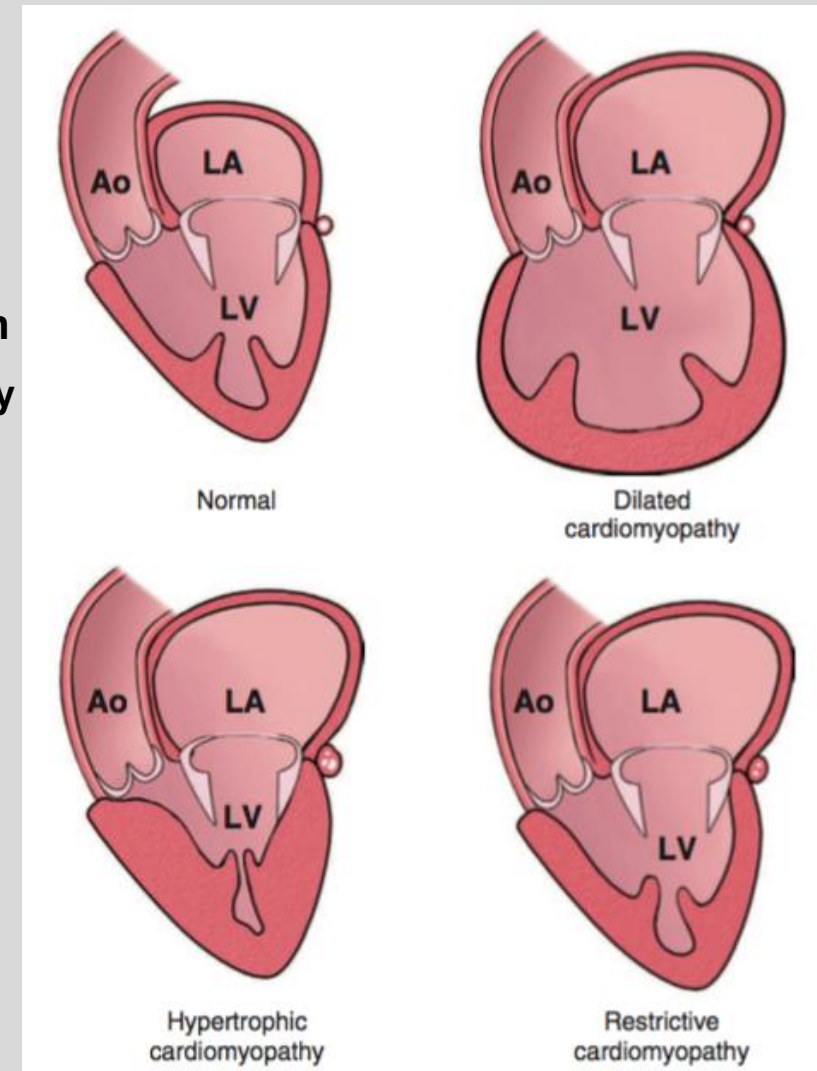
- **Generally structural** cardiac abnormalities manifesting as failure of myocardial performance
 - **Distinct from** impairment from **defined cardiovascular diseases** like hypertension, valvular disorders, and coronary artery disease
 - Etiology is **often** unknown **idiopathic**
- Common signs of heart failure include **shortness of breath**, unusual **fatigue**, and **swelling** in the legs, ankles, or feet caused by fluid buildup
 - Symptoms can develop slowly or start suddenly

Primary and secondary cardiomyopathies

- **Primary** cardiomyopathies involve mainly the heart
 - Can be **genetic** or **acquired**, such as acquired diseases like viral myocarditis, or anthracycline chemotherapy
 - Genetic causes include **mutations** in myocardial proteins of contraction, cell-cell contacts, or cytoskeleton (example, abnormal ion transport across cell membrane that can cause **dysrhythmias**)
- **Secondary** cardiomyopathies are a **component** of systemic or multi-organ disease processes, such as hemochromatosis or amyloidosis

Cardiomyopathy types

- **Dilated** Cardiomyopathy (DCM)
 - **90%** of cases
 - Abnormality = **Enlarged** chambers
 - Leads to **systolic contraction dysfunction**
 - Includes **Arrhythmogenic Cardiomyopathy**
- **Hypertrophic** Cardiomyopathy (HCM)
 - Abnormality = **Thickened** ventricular wall
 - Leads to **diastolic relaxation dysfunction**
- **Restrictive** Cardiomyopathy (RCM, next lecture)
 - **Least frequent**
 - Abnormality = **Stiff** myocardium
 - Leads to **diastolic relaxation dysfunction**



Compensatory mechanism effects

- Due to this, patient may be **asymptomatic early on**, but eventually **heart failure develops**
- **Angiotensin II** → ↑SVR → **afterload** becomes too high for **impaired LV**
- **Aldosterone** → ↑intravascular/EDV → fluid backs up → **pulmonary & systemic congestion**
- **Chronic** elevations in aldosterone and angiotensin II → cardiac **remodeling & fibrosis**
- With heart enlargement, **mitral/tricuspid regurgitation** may occur causing
 - Regurgitation of blood into the **LA** further **decreases** forward **stroke volume** during **systole** into aorta & systemic circulation
 - ↑**volume & pressure loads** in atria causing them to **dilate**; possible **atrial fibrillation**
 - Regurgitant volume returns to the **LV** during each **diastole** resulting in **greater volume load for LV** to pump out (dilation, LV **heart failure**)

DCM clinical features

- **Slow, progressive signs/symptoms of CHF**
 - **Signs** include cool extremities, **low arterial pressure**, **tachycardia**
 - Pulmonary venous congestion leads to auscultatory **crackles**
 - Left & upward displacement of **PMI**, **enlarged heart**
 - Third heart sound (**S3**) - blood rushing into a compliant, dilated, overloaded ventricle
 - **Normal** in **young** & athletes
 - **Abnormal** in **older** adults indicating **systolic heart failure**
- Complications: Mitral/Tricuspid **regurgitation**, atrial **fibrillation**, mural **thrombus/thromboembolism**
- Annual mortality high: 10-50% die usually from progressive cardiac failure or arrhythmia

DCM Pathogenesis

Peripartum associated:

- Occurs late in **pregnancy** or up to 6 months **postpartum** & may **recur**
 - Probably **multi-factorial** - **poorly understood** immune response, cytokine release, pregnancy associated hypertension, volume overload, nutritional deficiencies
- **50% regain normal** ventricular function
- **Risks** include advanced maternal **age**, African American **race**, **multiple gestations**

Genetic Causes:

- Familial in **30-50% of cases** - **AD** inheritance is most common
 - Mutation in *TTN* gene encoding **titan** may account for ~ 20% of cases
- Less commonly **X-linked**, **AR**, or **mitochondrial** inheritance
 - **X-linked DCM** linked to gene for **dystrophin** (links cytoskeleton to extracellular matrix) – **muscular dystrophy signs** after puberty to early **adult**
 - Deletions in **mitochondrial** genes → problems with FA oxidation or oxidative phosphorylation; typically **pediatric** manifestation

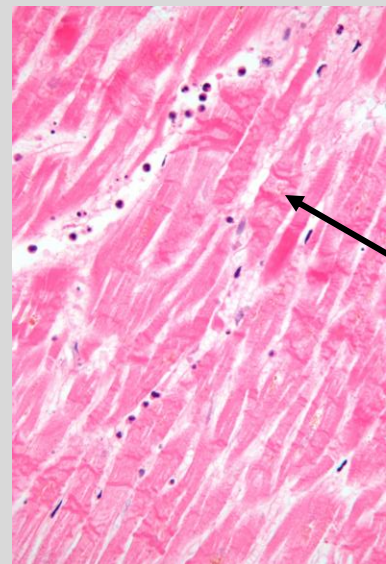
DCM pathogenesis

Toxin-Induced

- Strong association with **chronic excessive alcohol use**
 - Alcohol metabolite **acetaldehyde** = **direct toxin** to myocardium
 - **Ethanol (acetaldehyde)** → inhibition of mitochondrial oxidative phosphorylation and fatty acid oxidation → cell function declines
 - **Chronic** alcoholism → **thiamine** deficiency → **beriberi** - DCM & beriberi heart disease indistinguishable
 - Can be **reversible**: ventricular function improves with **cessation of alcohol**
- **Other toxins**: **Adriamycin** chemotherapy agents (**Doxorubicin**/Daunorubicin), cobalt, chronic **cocaine** use, **iron** overload - **hemochromatosis** (genetic or exogenous, ex. via transfusions)

Supraphysiologic Stress: Persistent **tachycardia**, **hyperthyroidism** lead to DCM

- Other conditions include **pheochromocytomas** (epinephrine), **cocaine** or vasopressor agents such as **dopamine** - **catecholamines** toxicity may cause multifocal myocardial **contraction band necrosis**
- Also, **intense autonomic stimulation** due to intracranial lesion or **emotional duress**
 - **Takotsubo** (“octopus pot”) **cardiomyopathy/ heartbreak syndrome** - left ventricular contractile dysfunction following extreme **psychological stress**
 - Left ventricular apex is most often affected leading to “**apical ballooning**”
 - Mechanism uncertain – maybe **calcium overload** or focal **vasoconstriction**



H&E stain: Contraction band necrosis

https://www.google.com/url?sa=t&source=web&rct=j&url=https%3A%2F%2Fen.wikipedia.org%2Fwiki%2FContraction_band_necrosis&ved=0CBkQjhxqFwoTCJj_5MaqnJYDFQAAAAAdAAAAABAY&opi=89978449

DCM treatment

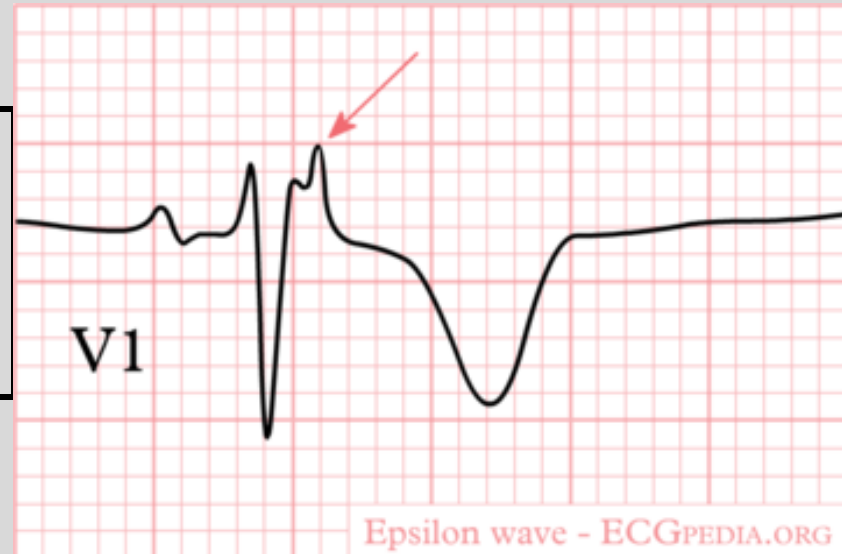
- Goals: Treat **early**, treat **underlying cause**, **symptom** relief, decrease backward vascular **congestion**, increase forward **CO**, prevent complications, prevent **thromboembolic** events, prevent and treat ensuing **arrhythmias**, improve long term survival
- Standard CHF **Rxs**: **Na⁺** restriction, **ACEI**/ARBs, **beta** blockers, **diuretics**, **digoxin**
- In those with poor LV function, anti- arrhythmia drugs may worsen a weak heart pump and lead to arrhythmias - **amiodarone** safest & does not carry a high risk of worsening heart failure
- **Anticoagulation** therapy may be for those prone to **thrombus** formation
- Implantable **cardioverter-defibrillator** (ICD) - prevents sudden cardiac death from dangerous heart rhythm
- **Cardiac transplant** may be the only definitive treatment
 - With no transplantation, 5 yr survival =74%; 10 yr = 55%
- 1/3rd of patients **spontaneously improve** after diagnosis

Arrhythmogenic Right Ventricular Cardiomyopathy (ARVC)

AKA Arrhythmogenic RV Dysplasia

- **Rare, progressive**, mostly sporadic
- Some familial forms occur: **AD** with variable penetrance - deletions of cardiac myocyte cell membrane **desmosome** junctional proteins → → **fibrosis** & myocyte loss w/ **fatty replacement**, or **aberrant signaling** between cells
 - Leads to **RV failure + rhythm disturbances**, esp. ventricular tachycardia and fibrillation with **sudden death** - associated with reentrant VT originating from the RV
 - Left sided involvement may occur
 - Can see **abnormal RV morphology** with non-invasive imaging
- Treatment usually involves **ICD** (implantable cardioverter defibrillator)
- **Naxos syndrome**: ARVC + **hyperkeratosis** of plantar/palmar surfaces associated with mutations in **plakoglobin** (desmosome-associated protein)

- Can pick this up on routine EKG
 - **Inverted T** waves in V1-V3
 - Possible **epsilon wave** = terminal QRS notch in V1
 - Indicates abnormal RV activation



Hypertrophic Cardiomyopathy (HCM)

- Hypertrophic obstructive cardiomyopathy aka idiopathic hypertrophic subaortic stenosis **incidence** 1 in 500 births worldwide
- **Poorly compliant** left ventricular myocardium leading to **abnormal diastolic filling**
 - **Intermittent** ventricular outflow **obstruction** (~1/3rd) - **diastolic dysfunction**; **systolic** function is usually **preserved**
- Unexplained LVH in **absence of external stimuli** (e.g., HTN or aortic stenosis)
 - Disease of **sarcomere proteins**, causing massive **primary** myocardial hypertrophy of ventricular wall without dilation → ↑ muscle mass existing **vasculature cannot support** → ↓ compliance & outflow **obstruction**, coronary arteries also narrow due to wall hardening → **myocardial ischemia**
- Heart is thick-walled, heavy, and **hypercontracting vs** flabby, hypocontracting heart of **DCM**

HCM etiology

- ~75% **familial** with **AD inheritance**; variable expression
- **100's** of known **mutations**, encoding for **sarcomere proteins**
 - Most are missense point mutations in 9 genes
 - **beta-myosin heavy chain (most common)**
 - cardiac troponins
 - myosin binding protein C
 - alpha-tropomyosin
 - associated w/ Friedreich Ataxia (often leads to severe heart complications)
- Sequence of events from mutations to disease poorly understood
- **Prognosis varies** widely, correlates with specific mutations

HCM clinical features

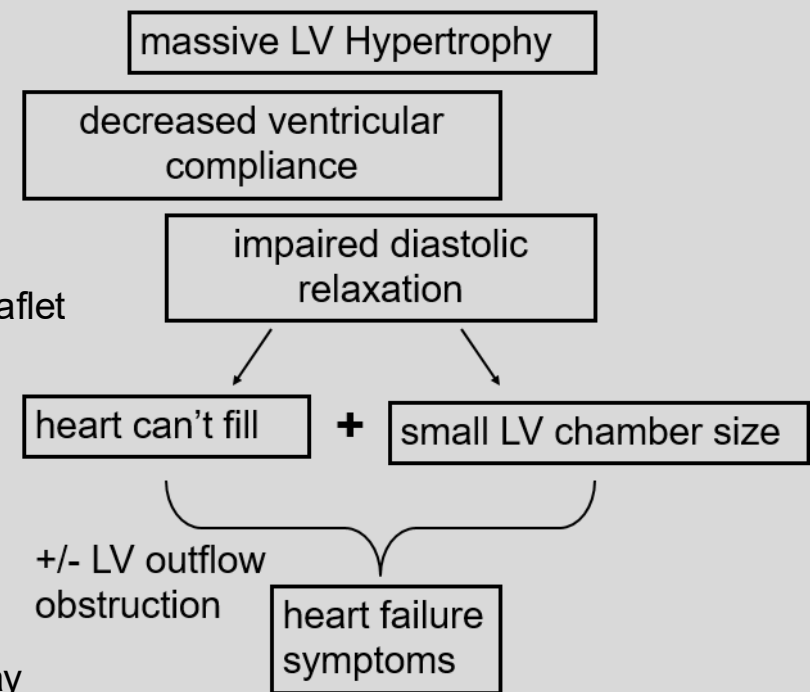
- From no to extensive limitations, **physical activity varies**
- Most do **not** have severe **outflow obstruction**
 - Most **symptoms** result from hypertrophy-induced **impairment in diastolic filling**
- Major clinical problems/complications: Mitral **regurgitation**, atrial **fibrillation** w/ mural **thrombus**, cardiac failure, ventricular arrhythmias, **syncope & sudden death**
 - **Most common** cardiac abnormality in **young athletes** who **die suddenly** during exertion (**ventricular arrhythmia**)
 - **Risks** = History of **syncope**, **family history** of sudden death, high risk **mutations**, extreme hypertrophy (**LV wall thickness** correlates with risk)

HCM: Clinical features mechanisms

	Hypertrophy	Obstruction
	Mechanism	
Dyspnea	diastolic impairment → ↑ LV, LA, PCW pressures	↑ systolic LV pressures during obstruction + possible MR
Angina	hypertrophy → ↑ muscle mass → ↑ O ₂ demand hypertrophy → squeezes down on small coronary branches in ventricular wall → impaired myocardial perfusion	Plus ↑ LV pressure during systole → ↑ wall stress → ↑ myocardial O ₂ demand
Light Headedness		standing → blood pools in legs → ↓ preload → ↓ LV return → ↓ LV chamber size → ↑ obstruction → transient ↓ in CO and cerebral perfusion
Syncope/ Arrhythmias Arrhythmias can worsen HCM symptoms like a. fib. causing loss of atrial kick so diastolic filling is further impaired and pulmonary congestion is worsened	Aberrant myocytes within conduction system → abnormal conduction & development of arrhythmias → syncope	↑ physical exertion → ↑ force of contraction worsens obstruction → transient outflow block → CO ↓ → syncope

HCM pathophysiology summary diagram - Rxs

- **Things that ↓ LV cavity** volume will bring anterior mitral leaflet & septum closer together
 - ↓ **venous return**
 - **Blood volume ↓**
 - **Positive inotropic drugs** ↑ force of contraction
 - So, this **promotes** obstruction
 - → Meds, surgery, ICD, avoid very heavy **exercise**, stay **hydrated**
- **Things that ↑ LV cavity** volume will push anterior mitral leaflet & septum apart
 - ↑ **venous return**
 - **Blood volume ↑**
 - **Negative inotropic drugs** ↓ force of contraction
 - So, this **reduces** obstruction
 - → Meds, surgery, ICD (for dysrhythmia), avoid very heavy **exercise**, stay **hydrated**



HCM clinical exam & treatments

Physical Exam

- **S4 heart sound** (LA contraction into **stiffened LV**)
- Outflow obstruction: **Harsh systolic ejection murmur**, crescendo, de-crescendo, caused by turbulent flow through the narrowed outflow tract; best heard at **left lower sternal border**

Screening

- **First degree relatives**
- **Serial echocardiograms** to note increases in hypertrophy during growth
- Should evaluate all HCM patients, even if asymptomatic - at risk for **sudden death**

Treatment

- Most do well with medications that relax the heart:
 - Standard therapy = **beta blockers**
 - Non-dihydropyridine Ca^{2+} **channel blockers** (verapamil)
 - Antiarrhythmic drugs such as **amiodarone**
- Avoid strenuous **exercise & drugs** that ↓ **preload** or ↑ **force of contraction**
- Implantable Cardioverter - Defibrillator (**ICD**): For patients at risk of sudden cardiac death (from dysrhythmias)
- Surgery: **Myomectomy** (more proven to hold) or septal **ablation** with alcohol